Digital Banking Industries

Enterprise Data Governance Framework Design & Implementation for Digital Banking Transformation

Created By: Felice Benita

Project Overview

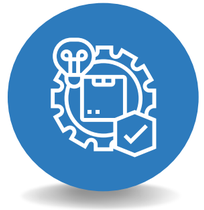


Project Objectives

The project aims to:

Strategic Objectives

✓ Establish enterprise-wide data governance
✓ Improve data quality
✓ Ensure regulatory compliance
✓ Define data ownership
✓ Improve decision-making
✓ Increase trust in data
✓ Enable AI/Analytics readiness



Scope of Data Governance

Business Domains

- Customer Data
- Account Data
- Loan Data
- Deposit Data
- Transaction Data
- Payment Data
- Digital Channel Data
- Fraud Data
- Risk Data
- Regulatory Reporting Data

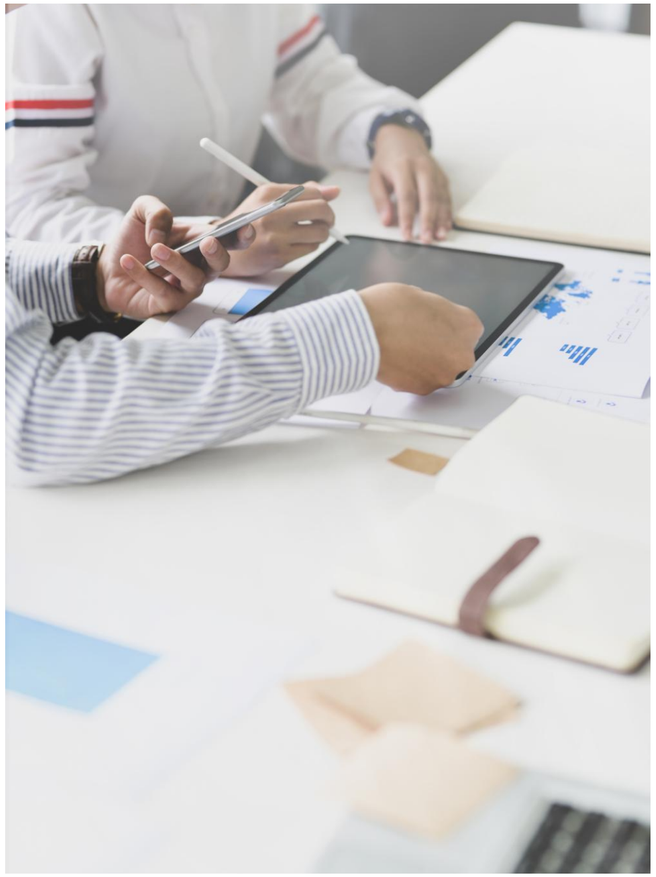


Business Context

A digital bank operates multiple data sources across:

- Mobile Banking Application
- Internet Banking
- Core Banking System
- Loan Origination System
- CRM
- Payment Gateway
- Fraud Detection System
- Data Warehouse
- Data Lake
- Regulatory Reporting Platform

Executive Summary



Data is the New Banking Asset

Digital Banking generates data from:

- 🏛️ Core Banking
- 📱 Mobile Banking
- 💳 Payments
- 💰 Lending
- 🛡️ Fraud Systems
- 📈 Analytics

However:

- Data ownership unclear
- Customer data duplicated
- Regulatory reporting inconsistent
- Data quality unmanaged

Result:

- ❌ Operational Risk
- ❌ Compliance Risk
- ❌ Poor Customer Experience
- ❌ Reduced Trust in Data

Current Challenges

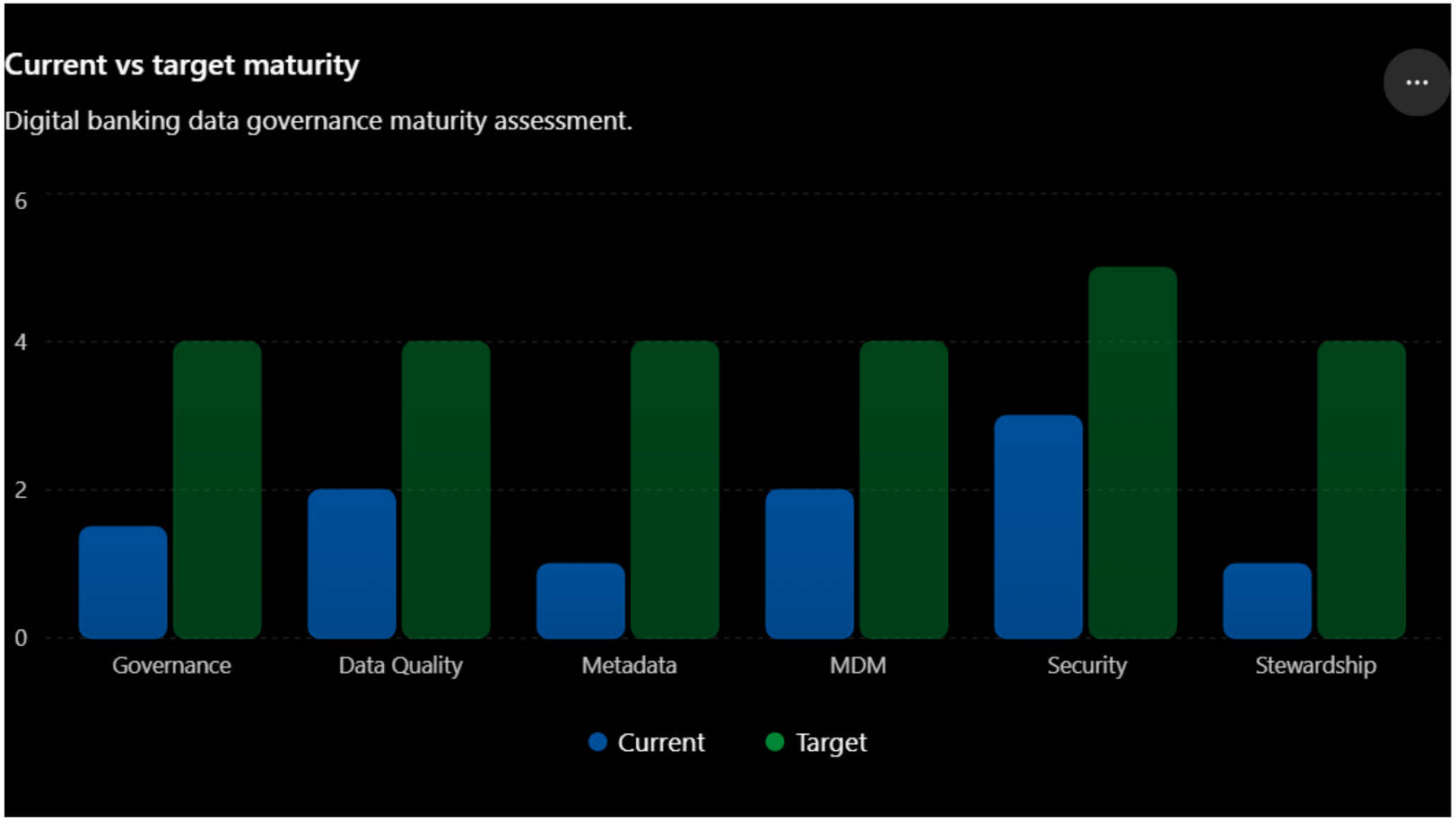
Problem	Impact
Data silos	Inconsistent customer information
Poor data quality	Incorrect reporting
Lack of ownership	Accountability issues
Regulatory compliance risk	OJK, BI, Basel, GDPR/PDPA exposure
Manual reporting	Operational inefficiency
Duplicate customer records	Poor customer experience
Uncontrolled access	Security risk

Current State Assessment (AS-IS)

Data Maturity Assessment

Area	Current Score	Target
Governance	1/5	4/5
Data Quality	2/5	4/5
Metadata	1/5	4/5
Master Data	2/5	4/5
Data Security	3/5	5/5
Data Lineage	1/5	4/5
Compliance	2/5	4/5

Current vs Target Maturity



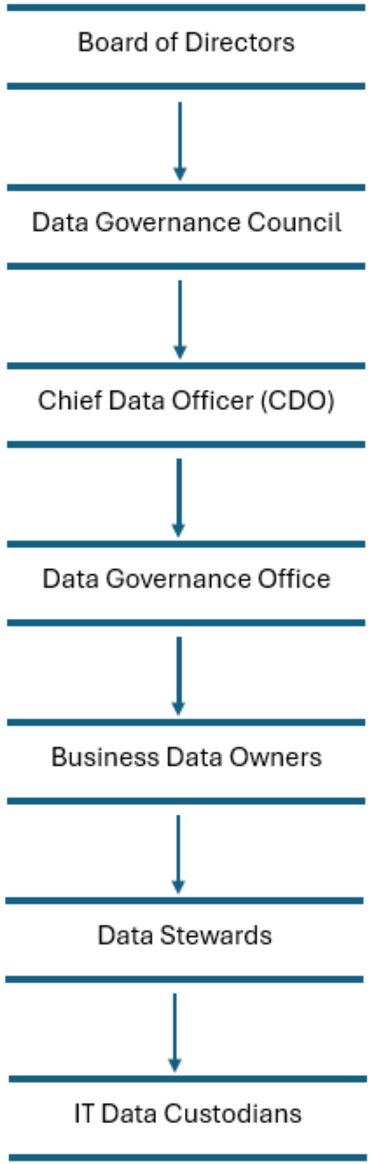
Current maturity level is fragmented and below industry benchmark.

Gap Assessment

Gap Analysis Matrix

Capability	Current State	Future State	Gap
Data Ownership	Undefined	Formal Owners	High
Data Quality Monitoring	Manual	Automated	High
Metadata Management	Spreadsheet	Enterprise Catalog	High
Data Lineage	Not Available	End-to-End Lineage	High
Data Policy	Fragmented	Enterprise Standard	High
Access Control	Inconsistent	Role-Based Access	Medium
Data Issue Management	Ad Hoc	Workflow Driven	High
Regulatory Reporting	Semi-Manual	Controlled Process	Medium

Target Operating Model (Federated)



Governance Model Type:

Federated Governance Model

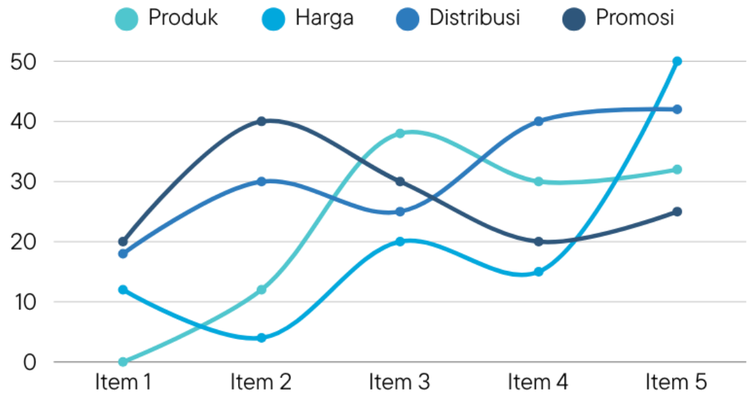
Recommended for Digital Banking because:

- Multiple business domains
- Strong regulatory oversight
- Shared accountability
- Scalable

Data Governance
Target Operating Model (TOM)

Link to: 📜 DATA GOVERNANCE TARGET OPERATING MODEL (TOM)





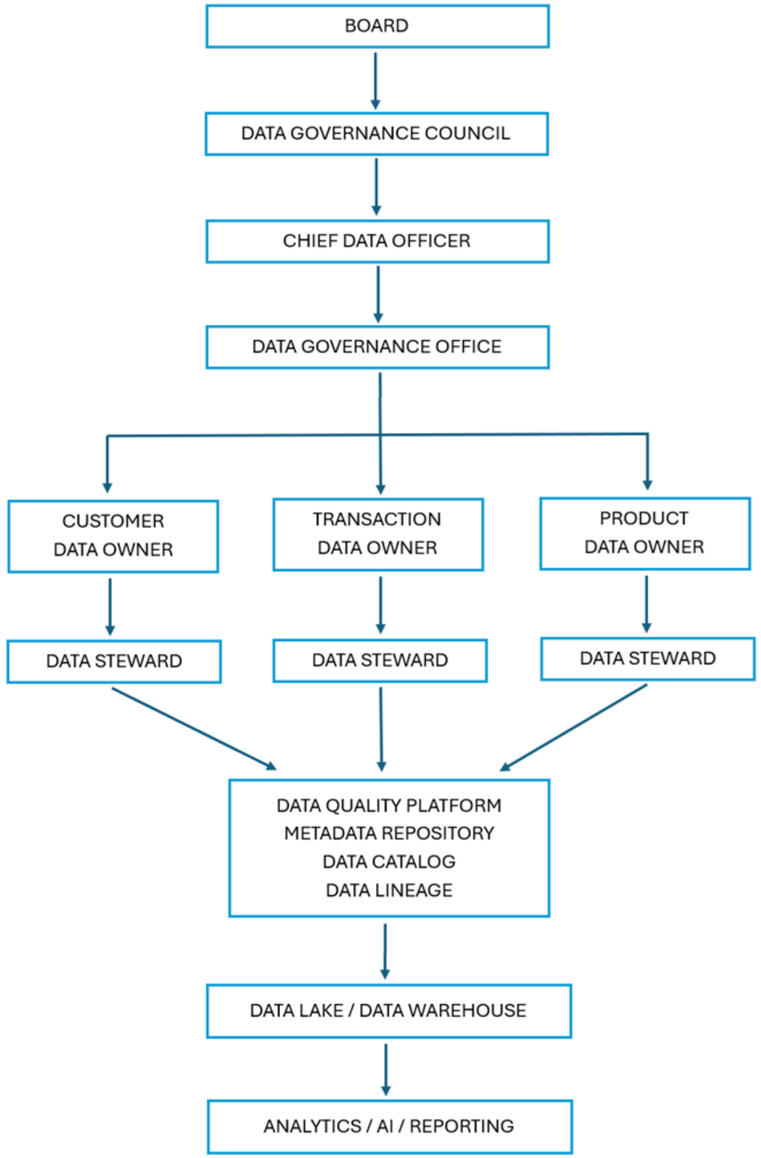
RACI Matrix

Activity	Data Owner	Data Steward	IT Custodian	DGO
Approve business data definitions	A	R	C	C
Define and maintain DQ rules	A	R	C	C
Monitor DQ and coordinate remediation	A	R	C	C
Maintain business metadata	C	R	C	A
Approve business data access	A	C	C	I
Provision or revoke technical access	C	I	R/A*	I
Resolve domain data issues	A	R	C	C
Escalate cross-domain issues	C	C	C	R/A*

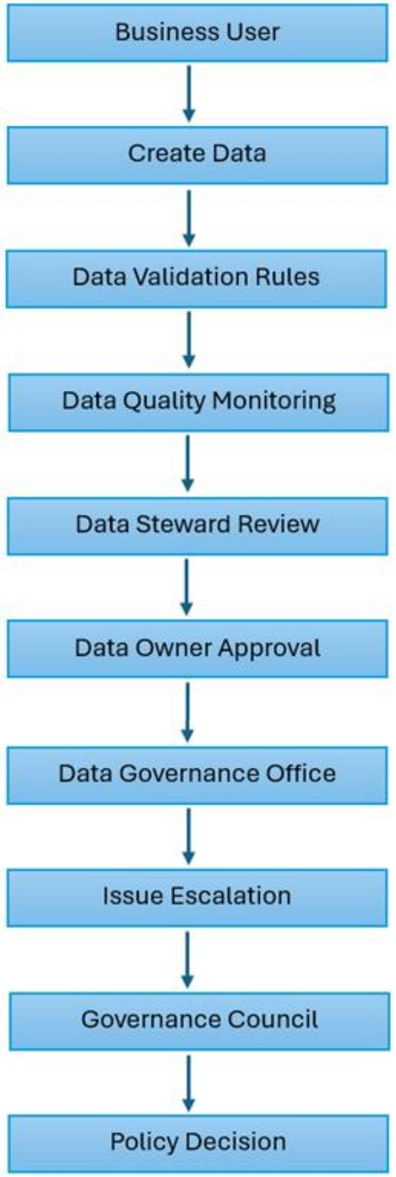
Legend: R = Responsible, A = Accountable, C = Consulted, I = Informed

*R/A means Responsible and Accountable may be assigned to the same function for certain operational activities. If the matrix requires one Accountable role per activity, explicitly assign a single decision owner and separate approval from execution.

Data Governance Architecture



Governance Flow Diagram



Data Governance Policy

Link to: 📜 DATA GOVERNANCE POLICY



Data Quality Management Framework

Critical Data Elements (CDE):

- Customer ID
- Account Number
- Mobile Number
- Transaction Amount
- Loan Outstanding

Data Quality Monitoring

KPI	Target
Completeness	>95%
Accuracy	>98%
Timeliness	>95%
Validity	>99%





Implementation Roadmap

Phase 1

Assessment

Deliverables:

- Current State Assessment
- Gap Assessment
- Stakeholder Mapping

Phase 2

Framework Design

Deliverables:

- Governance Model
- Policies
- Standards

Phase 3

Technology Enablement

Deliverables:

- Data Catalog
- DQ Dashboard
- Metadata Repository

Phase 4

Pilot Implementation

Domains:

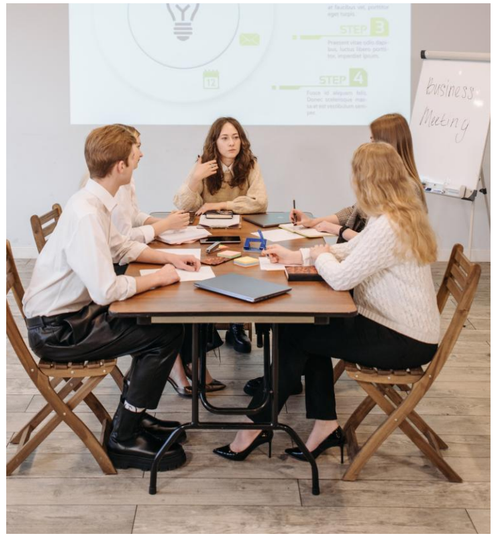
- Customer
- Transaction

Phase 5

Enterprise Rollout

Deliverables:

- Enterprise Governance
- KPI Dashboard
- Compliance Reporting
- Continuous Improvement



KPIs & Success Metrics

KPI	Before	Target	Success Criteria
Data Governance Maturity	Level 1–2	Level 4	Managed & Measurable Governance
Data Quality Score	82%	≥98%	Trusted Enterprise Data
Metadata Coverage	10%	≥95%	Complete Data Documentation
Data Lineage Coverage	5%	≥90%	End-to-End Traceability
Regulatory Findings	8 Findings	0 Critical Findings	Regulatory Compliance
Duplicate Customer Records	15%	<2%	Single Customer View
Report Reconciliation Effort	3–5 Days	<1 Day	Faster Reporting
Data Issue Resolution SLA	20 Days	≤10 Days	Efficient Issue Management

Conclusion & Recommendations

The bank should adopt a **Federated Data Governance Model supported by a dedicated Data Governance Office, enterprise-wide stewardship program, and modern data management capabilities**. By embedding governance into everyday business operations and leveraging enabling technologies, the organization can achieve trusted data, regulatory compliance, operational efficiency, and AI readiness, positioning itself for sustainable growth in an increasingly data-driven digital banking environment.

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THANK YOU

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